Parametric Design and Digital Fabrication of Temporary Pavilions for Resilient Historical Open Spaces

An Educational Experience in Parma

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The historic centers have maintained their structure almost unchanged throughout various eras, thanks to a conservative approach that preserves valuable architecture and urban systems. The preservation of pre-existing structures can maintain the identity of a place, but it may also limit the flexibility of urban spaces that evolve over time in response to changing climates and functions. The study aims to develop a systematic and holistic workflow for creating temporary architectures that enhance the climatic and functional resilience of historical open spaces. Several disciplines are brought together for project development, including historical and climate analysis, algorithmic design, fabrication, and digital prototyping. As both climate and cultural changes are occurring rapidly, it is essential to educate the younger generation of students on effective methods. The technologies of algorithmic design, combined with digital fabrication, enable the creation of customized solutions that are more effective than standardized construction models. This paper reports on a methodological experiment for developing temporary pavilions in the courtyard of the Cloister of San Sepolcro in Parma, Italy.

Keywords: *Algorithmic Parametric Design, Digital Fabrication, Laser Cutting, Prototyping, Temporary Architecture, Historical Spaces Space*

INTRODUCTION

Urban open spaces in historic centers are the result of a building stratification that has been consolidated over various eras. Typologies, methods, and building materials used in traditional architecture serve as valuable evidence of the past. In Italy, as well as in other countries, is preserved with great care and attention. The historic city faces the dual challenge of adapting to both climatic (Alcamo, 2012) (Kumar, 2021) and cultural changes, while simultaneously preserving its unique identity. The modifications allowed for open space designs are highly restrictive, necessitating the exploration of minimally invasive and temporary intervention strategies (Gherri, 2021).

The case study examines the courtyard of a religious building located in the city of Parma, Italy: the cloister of San Sepolcro. Previous studies analyzing the climate have shown that the San Sepolcro cloister courtyard experiences several critical climatic issues, particularly in terms of providing comfortable summer climate. Like many other historical cloisters, San Sepolcro is underutilized due to its secluded location and the absence of practical amenities for everyday use. This is mainly due to the fact that historical cloisters and courtyards are often overlooked as common public spaces, despite offering many thermal and functional advantages in the urban fabric. New infrastructure is instrumental in the readjustment of spaces. Permanent architectural interventions are prohibited; a low-impact approach to utilizing the courtyard is preferred. The courtyard typology, which is common in old town centers, exhibits similar climatic and functional discomfort characteristics. These findings can be applied to the entire typology.

The study aims to establish a workflow for designing and implementing temporary architecture that addresses climate adaptation, the repurposing of open spaces, and the relationship with building heritage. The available tools include algorithmic generative software and numerical control machines for digital fabrication. These tools can be used to systematize all data related to the identified themes, as well as for the development, prototyping, and production phases. The workflow was tested by Civil Engineering master's students, who had the opportunity to enhance their knowledge of both tool-related technologies and process-related methodologies.

BACKGROUND

The use of advanced design and construction processes is becoming increasingly prevalent in the field of architecture. These methods are expressed through architectural buildings, temporary structures, and street furniture. The benchmarks adopted for the morphogenesis of these architectures are primarily numerical and can be easily adapted to structural, climatic, or dimensional data of the context. Constructing qualitative relationships between visitors, new architecture, and pre-existing built environments, particularly in historical contexts, is a less immediate task.

OBJECTIVES

The study aims to enhance the resilience of open historical spaces by defining a holistic systemic workflow that creates a balance between various areas. These areas include the valorization of historical heritage, visitor experience, climatic adaptation, development of a repeatable building system for similar contexts, and scalability and adaptability of the project. Algorithmic analysis, design models, and digital fabrication technologies (Yuan, 2018) are the tools utilized to accomplish the research objectives.

Algorithmic design, which is used in compositional processes, defines complex relationships between the parts of architecture. A qualitative level of relationships, as opposed to a quantitative one, allows for the emergence of visual relationships and topological forms (Deleuze, 1994) (De Landa, 2000) rather than typological ones. The design structure of a space influences the relationship that visitors have with it. A temporary design structure can create new frames and viewpoints that highlight pre-existing elements in fragments. The solar radiation, which is partially filtered and partially obstructed by the loggia, creates both a sense of thermal comfort and an immersive experience.

Advanced technologies that work with numerical data are more easily utilized to obtain quantitative results. The effort to include non-quantifiable aspects broadens the scope and requires a holistic approach. The interplay of visual relationships through the fragmentation of the view, the creation of frames that incorporate the pre-existing architecture and portions of the sky, and the establishment of non-trivial analogies between the host courtyard and new construction are all factors that cannot be quantified but should be considered in the algorithm design process.

METHODOLOGY

The research methodology is divided into the following stages:

1. historical, environmental and climate analysis
2. definition of issues and strategies
3. choice of technologies, software and production tools
4. design morphogenesis

5. prototyping
6. construction of scale models

The historical analysis of San Sepolcro cloister aids in comprehending the identity of the location and its background. The climate analysis is conducted using simulation software such as Rhino, Grasshopper, and Ladybug (McNeel, R. & Associates, Rhinoceros, 2021). These data support previous studies that analyzed the microclimatic performance of the inner cloister using the Envi-MEt software. The analyses performed revealed an underutilization of the space in the courtyard inside the cloister of San Sepolcro, as well as critical issues related to thermal comfort during the summer season, in accordance with climate change. The UTCI and OTCA indices are used to evaluate thermal discomfort levels in the courtyard.

The strategy involves a temporary architectural intervention capable of reducing the summer thermal discomfort while also serving as infrastructure for visitors. The temporary architecture is developed using algorithmic design technology, the Rhino / Grasshopper / Ladybug generative software are adopted. Two sets of parameters are utilized in the development of compositional morphogenesis: those pertaining to climatic factors and those pertaining to the visitor's perspective. In the first scenario, the objective is to enhance the thermal comfort of the open spaces. In the second scenario, the goal is to create a dynamic interplay of perspectives that fragment and frame the pre-existing structure through the new construction. The prototyping phase involves a feedback process which first verifies the constructive feasibility of the architectural construction defined as described, then incorporates dimensional and quantitative changes in relation to the technical capabilities of the numerically controlled machines and the materials utilized. The design / production process culminates in the creation of a scale model, which serves to verify both the production process and the compositional aspects on a broader level.

Historical, environmental and climate analysis

The analysis of historical typology identifies the cloister of San Sapolcro (Melley, 2012) (Figure 1) as a recurring typology of an internal courtyard building, typical of ancient Italian centers. The San Sepolcro cloister features arcades on three sides and a spacious grassy courtyard. The original religious function is still maintained today; however, the spaces are now open to the public. Despite the potential for use as a public and communal space, the inner courtyard is rarely utilized.

The case study of San Sepolcro cloister is analyzed using simulation software such as Rhino, Grasshopper, and Ladybug. The same cloister underwent more detailed climatic analyses in a previous study (Touloupaki, Gherri, & Naboni, 2023), which utilized Envi-met software (ENVI_met, 2022). The climate simulations assessed the mean radiant temperature (MRT) values, wind speed, and air temperature, and evaluated the Universal Thermal Climate Index (UTCI) values, which revealed specific climatic issues resulting from the combination of the climatic zone and the typological conformation of the courtyard.

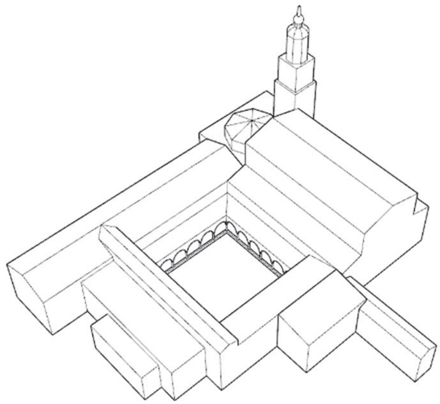


Figure 1
Closter of San
Sepolcro

Issues and strategies

The climate analysis highlights some critical points. Firstly, there is a high concentration of solar radiation in the upper internal facades of the courtyard. Secondly, there is a lack of both vertical and cross ventilation, which contributes to high air temperatures and discomfort during the summer season. These findings are supported by the UTCI and OTCA indexes. The design strategy involves utilizing modular construction systems to control shading, which reduces solar radiation on the courtyard floor while still allowing for vertical ventilation. Two design solutions are proposed, each with its own set of unique characteristics. The two structures are composed of distinct modules and share topological characteristics with the courtyard that surrounds them. Like the courtyard, they feature a perimeter frame that is more or less solid and a central opening. However, the compositional style differs typologically from the pre-existing structures and plays a contrasting role, which emphasizes both architectural languages.

The modules have a variable orientation, designed to optimize thermal performance. The various inclinations of the buildings in the courtyard create a dynamic interplay of viewpoints and framing, resulting in a visually engaging narrative experience for visitors. One of the two structures optimizes this aspect in an interesting way by identifying a point beneath the construction where the view of the cloister is fully visible despite being fully covered. The only fragmentation in the view is caused by the necessary modules. The viewpoint is at ground level, so visitors are encouraged to lie down on the lawn to experience this optical game.

Design morphogenesis

The concept moment already contains an overall setting of functional, climatic and cultural relations, and even the type of material, the construction system, the production logic in relation to a specific CNC machine. This is a systemic and holistic approach, which requires an in-depth dialogue phase between various disciplines even before setting up the work.

The digital composition of the two proposed projects derives from a process of morphogenesis and algorithmic analysis using Rhino / Grasshopper (Oxman & R., 2014) (Pugnale A. & M., 2007) software tools. This process involves five subsets:

- morphogenesis of the control surface
- morphogenesis of the construction system
- optimisation of the structure in relation to climatic and cultural parameters
- climate analysis using UTCI and OTCA indices
- parametric prototyping

The two projects A and B follow the same procedure. The control surface and analysis part undergo the same morphogenesis process. Differences are evident in the morphogenesis of the two construction systems, resulting in project-specific parametric prototyping. This study focuses on developing algorithms for building systems and parameterizing the prototyping phase. The study also includes the realization of a scale model. However, the climate aspect, including both morphogenesis and analysis, will be addressed in a separate study.

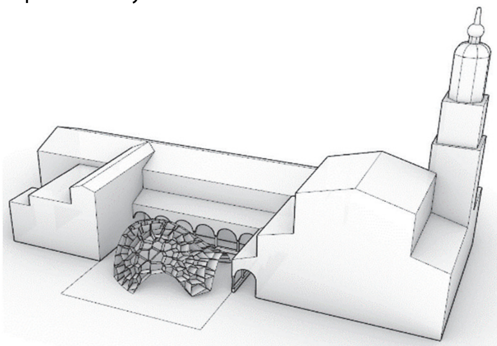


Figure 2
Project A in the cloister of San Sepolcro

In project A (Figure 2), the construction system is defined by dividing the control surface into irregular hexagons using a Voronoi algorithm. The extrusion of the different Voronoi segments on the surface creates a modular composition that is enclosed on the perimeter and open on both sides of each module. The composition of each module, as defined, has a topological structure similar to that of a court, with an opening on a closed perimeter. The perimeter enclosing the elements of each module has differently oriented positions in space, ranging from vertical to oblique. This arrangement is designed to optimize shading and consequent thermal comfort in relation to solar radiation.

The orientation of elements derives from a biomimicry process (Benyus, 2002) (Pawlyn, 2016). The composition of the Opuntia ficus-indica (Opuntia ficus-indica) (Anderson, 2001) (Figure 3) comprises vertically oriented and inclined flat elements, which bring shade to the body of the plant, but at the same time do not prevent the passage of vertical air. This strategy helps the plant to withstand the extremely hot and arid climate of its natural habitat. The same strategy can be effectively applied to an installation within a courtyard, providing shading without obstructing vertical ventilation. This is particularly important in the case of a courtyard building with a closed conformation, where vertical ventilation is the only option available.

Figure 3
Opuntia ficus-indica. Acatlán, Hidalgo, México,2013. Diego Delso.

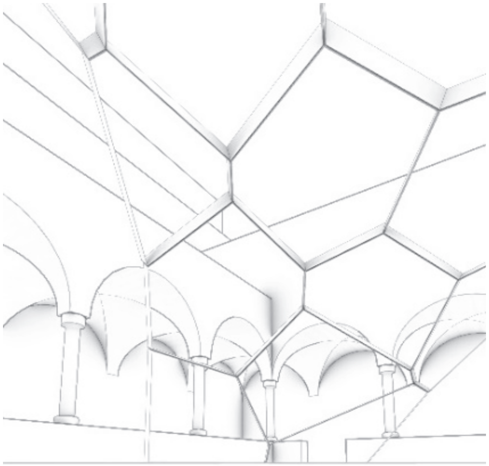


Figure 4
The vision of the cloister of St. Sepulchre through the modular structure of project A. Perspective game for the visitor.

The design incorporates compositional analogies with pre-existing structures and biomimicry climatic strategies inspired by cacti that thrive in extreme climates. A third objective is to establish a visual relationship between the visitor and the architecture. The temporary structure serves as both a closure and an opening to the surrounding context, which is defined by the building that borders the courtyard.

This game of perspectives involves multiple points of view that result in varying percentages of closure and aperture, with a specific point where the maximum visual opening is achieved. From one vantage point on the courtyard floor, the historic buildings are fully visible, while the temporary structure appears as a sparse grid, with each module serving as a frame (Figure 4). The visitor is invited to lie down on the grassy floor at the indicated point for the proposed immersive experience.

The described set of climatic and cultural characteristics is achieved through the implementation of algorithmic rules that guide the extrusion of modules towards a vanishing point. The amplitude of extrusion, however, is different for each module and is generated through morphogenesis optimization (Wortmann, 2017) (Rutten, 2010) from climatic parameters.

The construction system for project B (Figure 6) is defined by discretizing planar triangular modules of the control surface. Each module includes a triangular hole, which is created by offsetting the shape of the module itself. The size of the hole is determined by the morphogenesis of climatic parameters in order to optimize the summer thermal comfort of the courtyard spaces. In this project, the modules have a topological relationship with the courtyard's composition. They are delimited by a perimeter that delimits an opening.

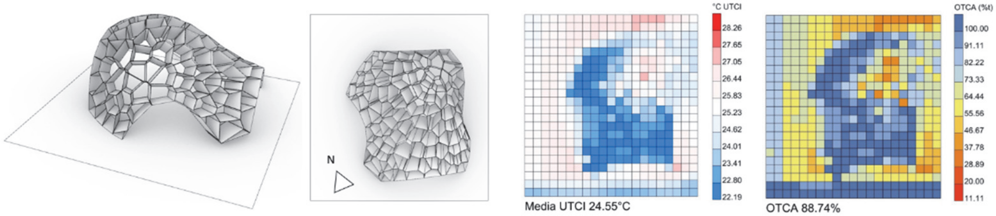


Figure 5
Climate analysis of project A. UTCI and OTCA

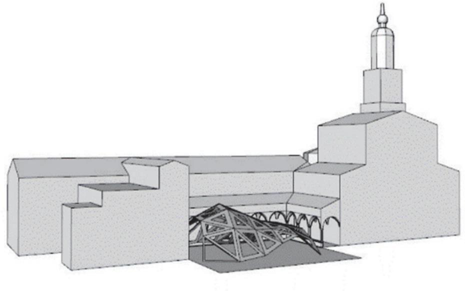


Figure 6
Project B in the cloister of San Sepolcro

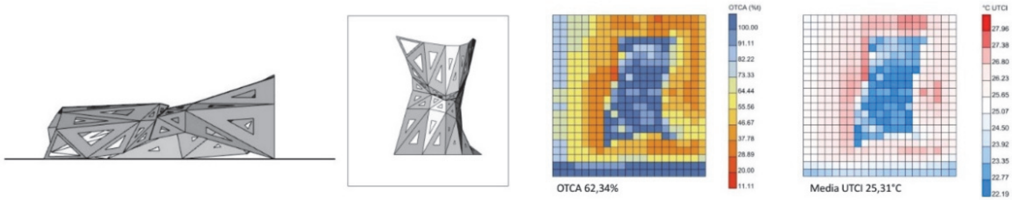


Figure 7
Climate analysis of project B. UTCI and OTCA

The alternation of empty and full spaces, created by the varying sizes of the modules and openings, results in a visual fragmentation of the area. As visitors cross the courtyard, they experience multiple perspectives of the same context.

Prototyping

The prototyping phase includes two moments, the first corresponds to the development of the algorithm and the second involves the scaling of the model in relation to the characteristics of the available CNC machines and the material. The CNC machine chosen is the laser cutter, the material for the scale model is corrugated cardboard.

For project A, the algorithmic process for prototyping involves combining all sides of each module into a single surface, which is then unrolled on the XY plane. As many surfaces are obtained as there are modules. An identification number marks the module and its corresponding unrolled surface. To close the surface, a flap is created on the corresponding side and folded over. The laser cutter requires a two-dimensional drawing with linear vector elements for both cutting and incisions. Therefore, the surfaces of the ashlars obtained as described are divided into two groups of linear vector elements: one for cutting, which corresponds to the external margin, and the other for incisions, which corresponds to the folding points (Figure 8). The cutting and incision work takes place on the extrados of the ashlar blocks (Figures 9, 10).

The size and number of modules, as well as the range of extrusion depth, are regulated based on the dimensional limits of the formats, the thickness of

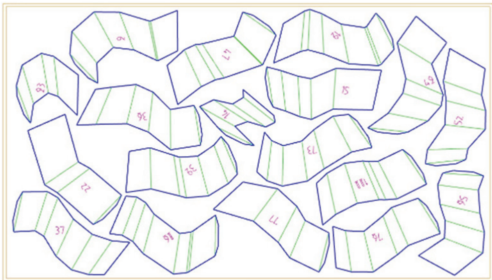


Figure 8
Project A. Nesting

Figure 9
Project A. Laser-cut modules

Figure 10
Project A. Modules

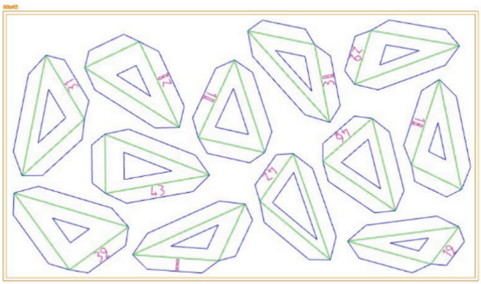


Figure 11
Project A. Nesting

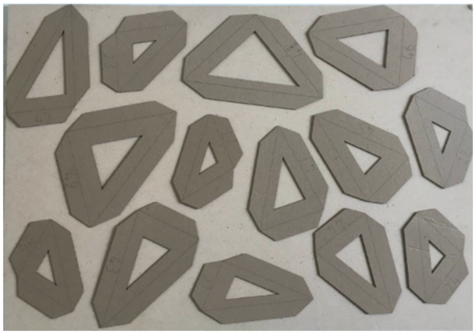


Figure 12
Project B. Laser-cut nesting

the material, and the cutting plane dimensions of the available laser cutter.

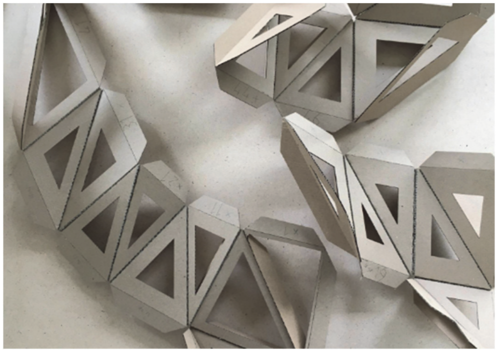
Project B involves planar modules that are joined together on three sides. In this case, the prototyping algorithm transfers the surfaces of the numbered triangular perforated modules onto the XY plane. These modules begin with multiple orientations and are then arranged into a planar organization. The modules are connected by aligning and bringing together the sides that coincide with each other. To complete the task, you must create tabs with parametric dimensions on all three sides of each module.

The surfaces of the modules are arranged in vector lines on two layers to distinguish between the cutting and engraving areas for the laser cutter (Figure 11). The prototyping phase, although it occurs at the end of the process, requires a complete revision of the initial project setup. Prototyping (Figure 12, 13) is an active phase of the project that involves more than just translating compositions into production schemes. It involves modifying the project itself to optimize dimensions and quantities, as well as improve assembly techniques. It is essential to consider the construction process and the subsequent logic of prototyping during the initial concept phase.

The nesting process (Figure 8, 11) for both projects optimize the organization of the modules in the material format. The production takes place through a laser cutting process of the cardboard. The assembly of the numbered modules returns the models on a reduced scale.

RESULTS

The research has resulted in a workflow for two building systems for temporary architecture that incorporate cultural aspects such as the relationship with historical heritage, interaction with visitors, functionalization of historical open spaces, climatic adaptation, and realisability through automated digital fabrication processes. The workflow, based on algorithmic processes, allows for a large family of solutions based on input parameters. This approach



offers the advantage of generating various design variants that can be quickly adapted to the dimensional or functional differences of other historic open spaces and courtyards.

The realization of the two small-scale models (Figure 15, 16) demonstrates the feasibility of the full-scale models. In fact, the process is identical.

CONCLUSIONS

Adopting a process that incorporates both quantitative and qualitative multi-criteria leads to the development of holistic and systemic solutions. Algorithmic design and digital fabrication are valuable tools for creating temporary architecture that enhances historical open spaces, such as courtyards or squares. As part of a systemic approach,

Figure 13
Project B. Modules

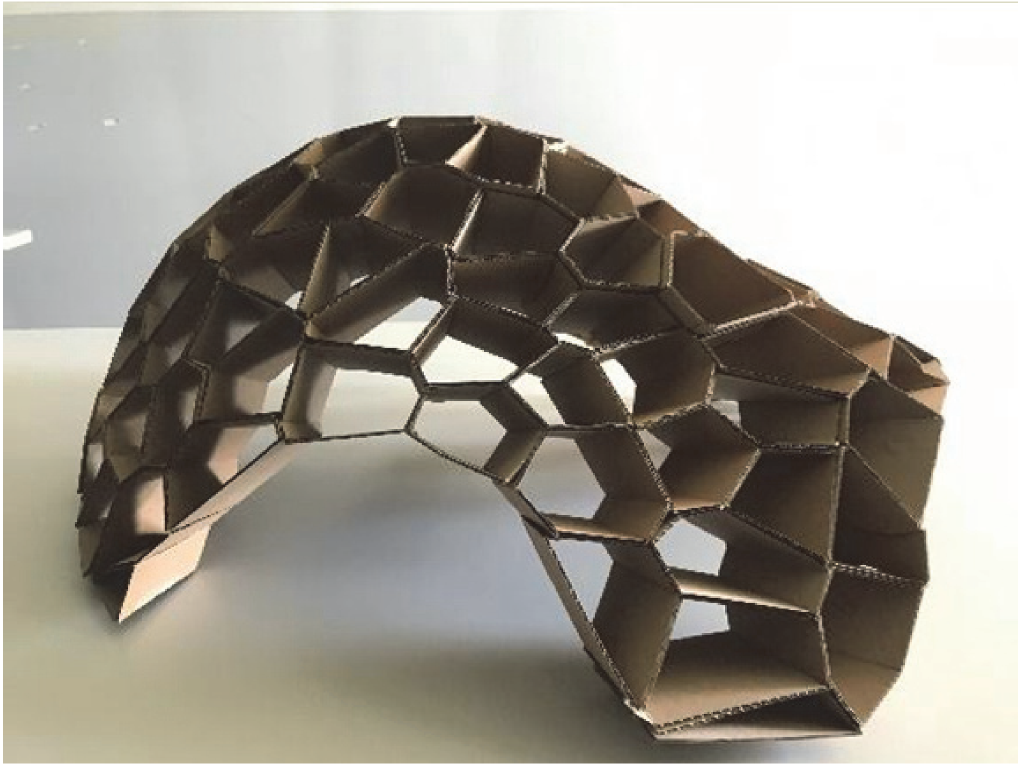
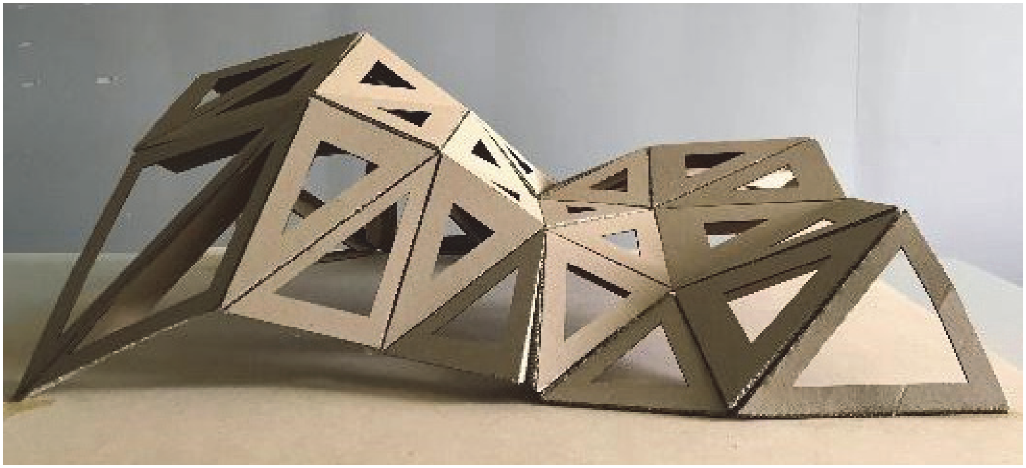


Figure 14
Project A. 1:30 scale prototype

Figure 15
Project B. 1:30 scale prototype



they can manage cultural aspects of the relationship with pre-existing architectures, experiential aspects of the relationship with visitors, and aspects of comfort in relation to climate. The result of all this is a resilience of historical open spaces.

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